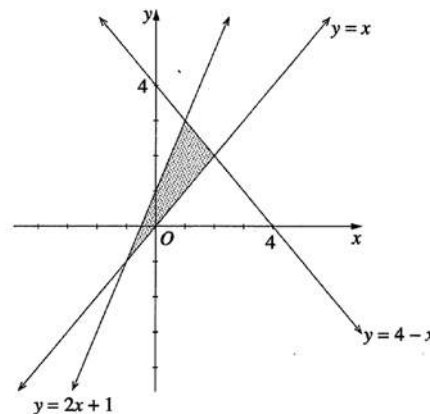


1. The region enclosed by $y = 4 - x$, $y = x$ and $y = 2x + 1$ is shaded in the diagram below. Which of the following defines the shaded region?

- A. $y \leq 2x + 1, y \leq 4 - x, y \geq x$
- B. $y \geq 2x + 1, y \leq 4 - x, y \geq x$
- C. $y \leq 2x + 1, y \geq 4 - x, y \geq x$
- D. $y \geq 2x + 1, y \geq 4 - x, y \geq x$



1 (A)

Use the point $(0, 0)$ to test the inequality.

$$y = 2x + 1$$

$$0 \leq 2 \times 0 + 1$$

$$y = 4 - x$$

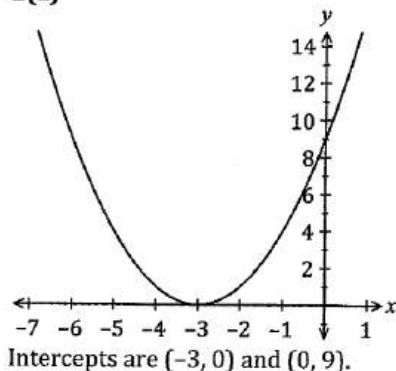
$$0 \leq 4 - 0$$

$$\therefore y \leq 2x + 1, y \leq 4 - x, y \geq x$$

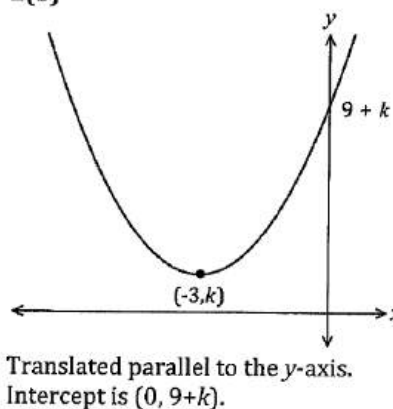
2. Sketch the curve of the following equations on separate diagrams.

- a. $y = (x + 3)^2$
- b. $y = (x + 3)^2 + k$, where k is a positive constant. Show on each sketch the coordinates of each point at which the graph meets the axes and any turning points.

2(a)



2(b)

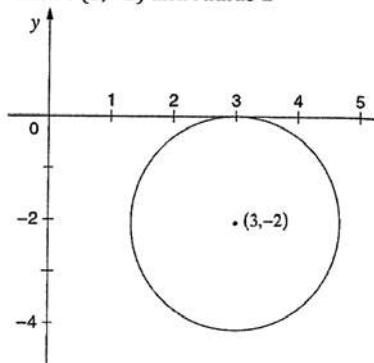


3. Sketch the graph of $(x - 3)^2 + (y + 2)^2 = 4$.

3

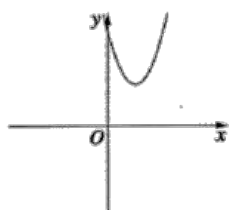
Equation is a circle.

Centre $(3, -2)$ and radius 2

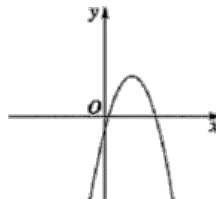


4. Which diagram best shows the graph of the parabola $y = 3 - (x - 2)^2$?

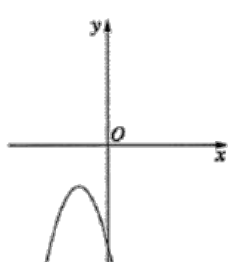
A.



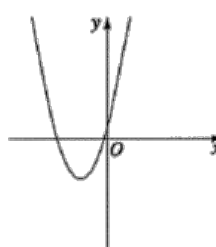
B.



C.



D.



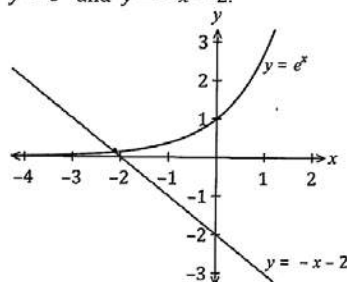
4 (B)

Parabola that is concave down with a turning point of (2, 3).

5. How many solutions exist for x of the equation $e^x + x + 2 = 0$? Hint: Draw graphs.

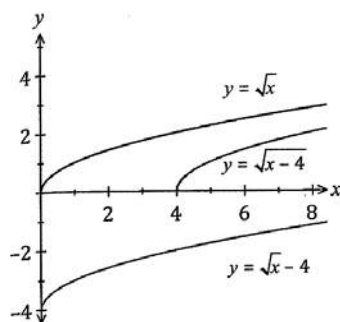
5

Draw the graphs of $y = e^x$ and $y = -x - 2$.



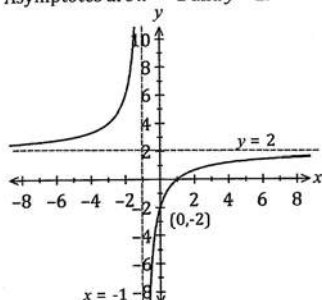
$\therefore$ There is one solution for $e^x + x + 2 = 0$
(point of intersection)

6. Sketch the following graphs on the same number plane: $y = \sqrt{x}$, $y = \sqrt{x-4}$, $y = \sqrt{x} - 4$



7. Sketch the graph of $f(x) = 2 - \frac{4}{x+1}$. Label all axis intercepts. Label each asymptote with its equation.

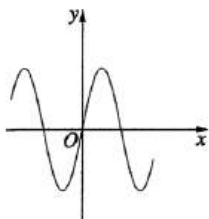
Intercepts (0, 2) and (1, 0).
Asymptotes are $x = -1$ and $y = 2$.



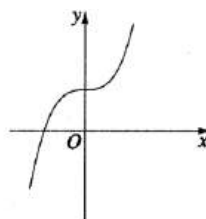
Transformations of Functions Topic Test 3

8. Which diagram shows the graph of an odd function?

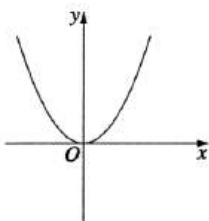
A.



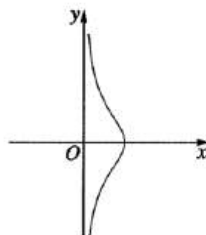
C.



B.



D.



8 (A)

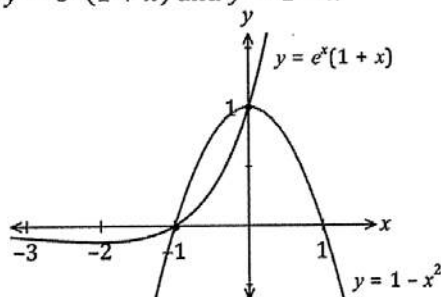
Odd functions have point symmetry.

9. The diagram shows the graph of $y = e^x(1+x)$. How many solutions are there to the equation $e^x(1+x) = 1-x^2$?

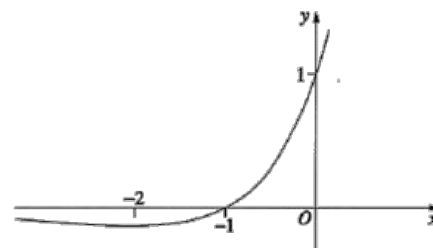
- A. 0
- B. 1
- C. 2
- D. 3

9 (C)

Draw the graphs of
 $y = e^x(1+x)$ and $y = 1-x^2$



There are 2 points of intersection.



10. a. Find the domain and range for the function $f(x) = \sqrt{9-x^2}$.

b. On a number plane, shade the region where the points (x, y) satisfy both inequalities $y \leq \sqrt{9-x^2}$ and $y \geq x$.

10(a)

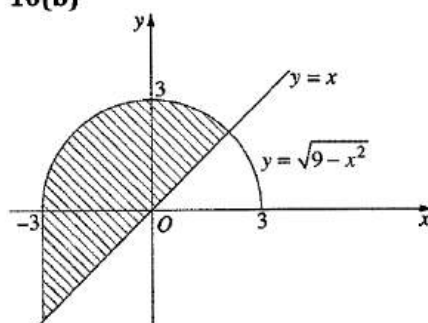
Domain $9-x^2 \geq 0$

$$x^2 \leq 9$$

$$-3 \leq x \leq 3$$

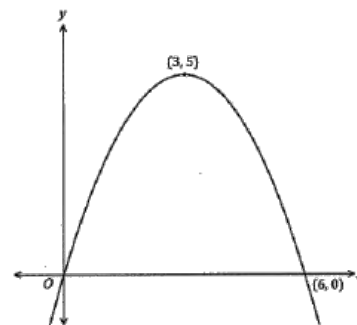
Range $0 \leq y \leq 3$

10(b)

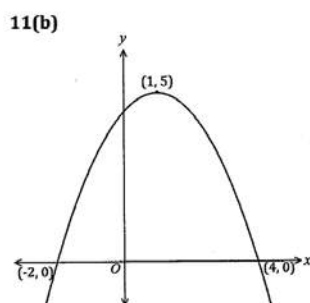
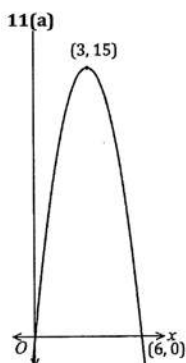


Transformations of Functions Topic Test 3

11. A sketch of the curve $y = f(x)$ is shown. The curve passes through the origin O and through the point $(6, 0)$. The maximum point on the curve is $(3, 5)$. Sketch the curve with the following equations, clearly showing the coordinates of the maximum point and where the curve crosses the x -axis.



- $y = 3f(x)$
- $y = f(x + 2)$



12. Given that $f(x) = (x^2 - 6x)(x - 2) + 3x$:

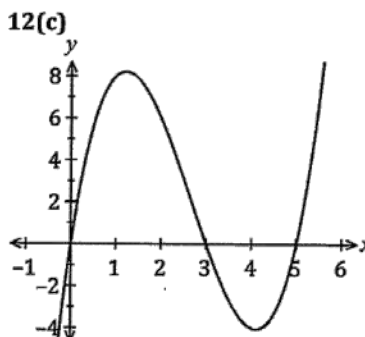
- Express $f(x)$ in the form $x(ax^2 + bx + c)$, where a, b and c are constants.
- Hence factorise $f(x)$ completely.
- Sketch the graph of $y = f(x)$, showing the coordinates of each point at which the graph meets the axes.

12(a)

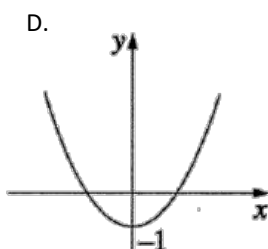
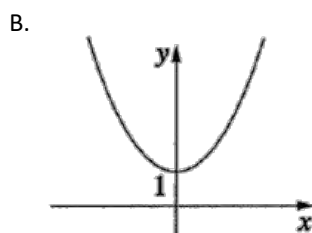
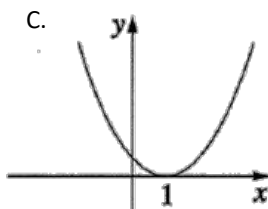
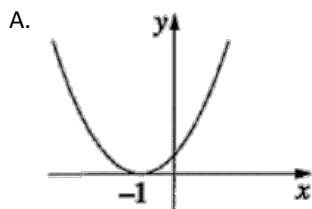
$$\begin{aligned} f(x) &= (x^2 - 6x)(x - 2) + 3x \\ &= x[(x - 6)(x - 2) + 3] \\ &= x(x^2 - 8x + 15) \end{aligned}$$

12(b)

$$\begin{aligned} f(x) &= x(x^2 - 8x + 15) \\ &= x(x - 5)(x - 3) \end{aligned}$$



13. Which graph best represents $y = (x - 1)^2$?

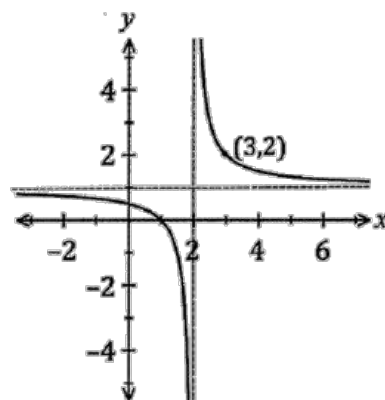


13 (B)
The graph $y = (x - 1)^2$ is the graph of $y = x^2$ translated to the right by 1 unit.

14. The diagram below shows the graph of $y = f(x)$.

Which of the following statements is false?

- A. The horizontal asymptote is $y = 1$.
- B. The curve is continuous.
- C. The curve is concave up for $x > 2$.
- D. The equation of the function is $y = \frac{x-1}{x-2}$.

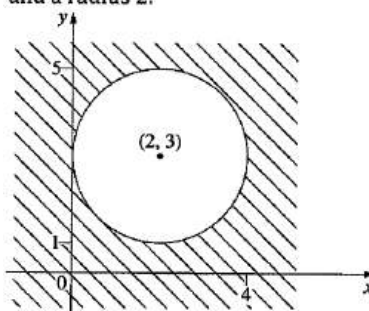


14 (B)

The curve has a point of discontinuity at $x = 2$.

15. Sketch the region of $(x - 2)^2 + (y - 3)^2 \geq 4$.

15
Exterior of a circle centre at $(2, 3)$
and a radius 2.



16. a. Factorise completely $x^3 - 4x$.

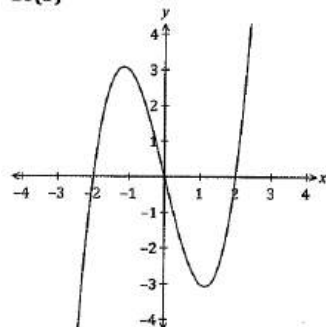
- b. Sketch the curve with equation $y = x^3 - 4x$, showing the coordinates of the points where the curve crosses the x -axis.
- c. Sketch the curve with the equation $y = (x - 1)^3 - 4(x - 1)$, showing the coordinates of the points where the curve crosses the x -axis.

16(a)

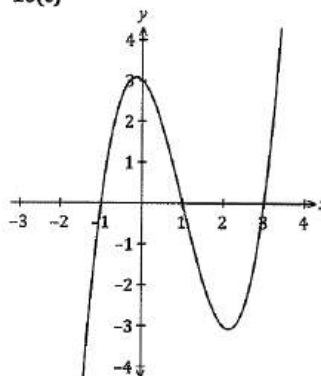
$$x^3 - 4x = x(x^2 - 4)$$

$$= x(x + 2)(x - 2)$$

16(b)



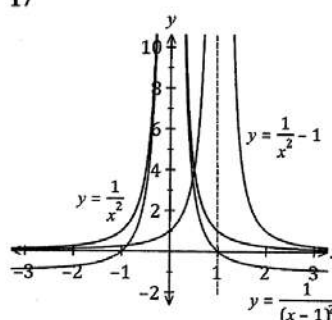
16(c)



17

17. Sketch the following graphs on the same number plane:

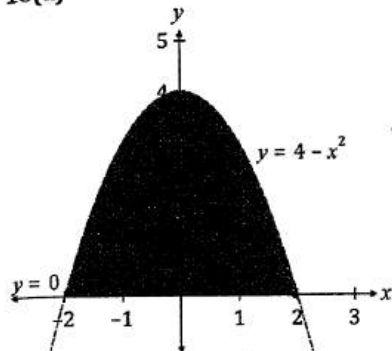
$$y = \frac{1}{x^2}, y = \frac{1}{(x-1)^2}, y = \frac{1}{x^2} - 1.$$



Transformations of Functions Topic Test 3

18. a. Sketch the region defined by the intersection of $y < 4 - x^2$ and $y > 0$.
b. Hence write the solution to $4 - x^2 > 0$.

18(a)



18(b)

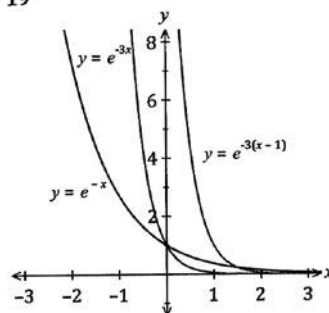
$$4 - x^2 > 0$$

Points of intersection: $(-2, 0)$ $(2, 0)$

$$\therefore -2 < x < 2$$

19. Sketch the following graphs on the same number plane:
 $y = e^{-x}$, $y = e^{-3x}$, $y = e^{-3(x-1)}$.

19

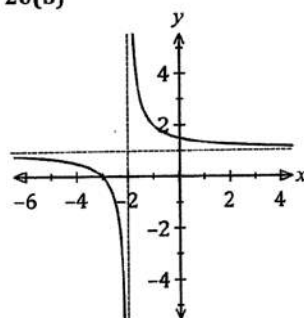


20. a. Show that the function $y = \frac{x+3}{x+2}$ can be written as $y = 1 + \frac{1}{x+2}$.
b. Hence sketch the graph of $y = \frac{x+3}{x+2}$ showing all the asymptotes.

20(a)

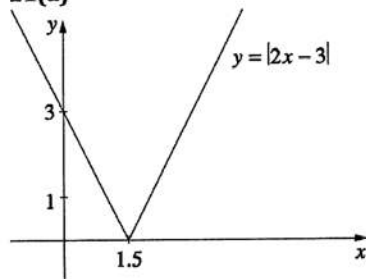
$$y = \frac{x+3}{x+2} = \frac{x+2+1}{x+2} = 1 + \frac{1}{x+2}$$

20(b)



21. a. Sketch the graph $y = |2x - 3|$.
b. Using the graph from part (a), or otherwise, find all values of m for which the equation $|2x - 3| = mx + 1$ has exactly one solution.

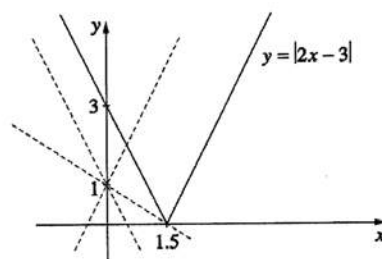
21(a)



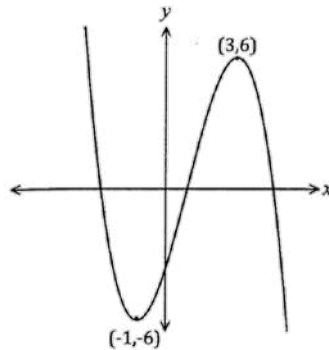
21(b)

From the graph

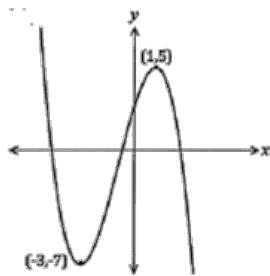
$$m < -2 \text{ or } m \geq 2 \text{ or } m = -\frac{2}{3}$$



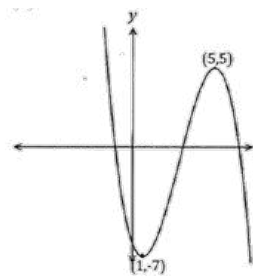
22. The diagram shows the graph of $y = f(x)$. Which of the following shows the graph of $y = f(x + 2) - 1$?



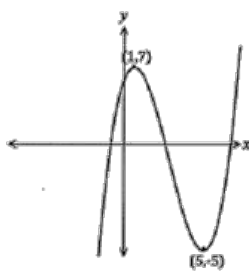
A.



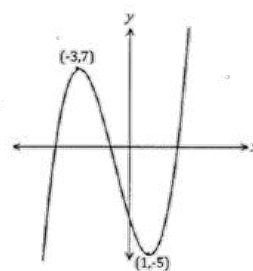
C.



B.



D.



22 (A)

